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Anonymous authors
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Summary

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Contribution(s)

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Abstract

1 TODO

2 1 Introduction

3 2 Position

4 Short summary of position

5 3 Definition

6 We work over a single, never-resetting stream of experience and assume no Markov, episodic, or
7 stationarity structure. Rewards lie in $[0, R_{\max}]$, and we measure value over an H -step look-ahead.
8 The agent is capacity-bounded: it represents policies and beliefs within a fixed budget of bits. We
9 seek a definition of a learning measure the agent can compute from its own state, the condition under
10 which it is trustworthy, and the limit beyond which it cannot be.

11 **Definition 1** (Stream, policy, value). A *history* is $h_t = (o_0, r_0, a_1, \dots, a_t, o_t, r_t)$, with observations
12 o_s , rewards $r_s \in [0, R_{\max}]$, actions a_s ; the underlying law is μ . A *policy* π maps histories to action
13 distributions, with value $f_t^\pi = \mathbb{E}_{\mu, \pi}[\sum_{s=t+1}^{t+H} r_s \mid h_t]$.

14 **Definition 2** (What the learner could have done). The learner represents policies within K_L bits;
15 write Π_{K_L} for this representable set. The *within-capacity optimum* is $f_t^{K_L} = \sup_{\pi \in \Pi_{K_L}} f_t^\pi$, and
16 the *structural gap* to the global optimum is $\Delta_t^{\text{str}} = f_t^* - f_t^{K_L} \geq 0$. We compare the agent to $f_t^{K_L}$,
17 the best it could itself express, and keep Δ_t^{str} visible as the irreducible price of being small.

18 **Definition 3** (Learner and examiner). An agent is a pair $(\mathcal{L}, \mathcal{E})$. The *learner* $\mathcal{L}$ emits the policy
19 $\pi_t \in \Pi_{K_L}$ it acts with, and learns only about that policy. The *examiner* $\mathcal{E}$ is a *predictive model*: from
20 the history it outputs an *aspiration* V_t^{hi} and a *guarantee* V_t^{lo} , with $V_t^{\text{hi}} \geq V_t^{\text{lo}}$. Setting V_t^{hi} requires
21 forecasting the value of policies the learner did not run, so the examiner must learn about a whole
22 family of behaviours, not the one acted. In the spirit of general value functions (??), temporally
23 abstract predictions (?), and off-policy prediction (?), V_t^{hi} and V_t^{lo} are value predictions the agent
24 learns from its own stream, and the examiner's *inductive bias*, the model class $\mathcal{M}_E$ of environments
25 it considers possible, the concentration rule, or the prior of its predictors, is what lets it generalize
26 to behaviours it has not tried. A Bayesian posterior over a model class is one instance, not the only
27 one (?).

28 **Definition 4** (Examiner regret). $\rho_t = V_t^{\text{hi}} - V_t^{\text{lo}} \geq 0$, with $\mathcal{R}_T^{\text{ex}} = \sum_{t=1}^T \rho_t$. It depends only on the
29 history and the examiner; it never refers to μ or to any optimum.

30 4 A Worked Case: The Examiner in a Bandit

31 The simplest world in which the definitions become concrete has no state at all. A multi-armed
32 bandit is the stream of Section ?? with $|\mathcal{S}| = 1$ and $H = 1$. We use it for two reasons: it is the

33 standard sanity check for any value-based definition, and it makes “a behaviour the learner never
34 tried” as transparent as possible, namely an arm it never pulled.

35 **Setting.** There are N arms; pulling arm i returns a reward $r \in \{0, 1\}$ with unknown mean μ_i , so
36 $R_{\max} = 1$. Every arm pays $\frac{1}{2}$ except a single *special* arm with $\mu_{i^*} = \frac{1}{2} + \epsilon$, and the agent acts
37 on one never-resetting stream. Write $\hat{\mu}_i$ for the empirical mean of arm i , n_i for its pull count, and
38 $c_i = \sqrt{2 \ln t / n_i}$ for the confidence half-width, with c_i at its maximum for an arm never pulled.

39 **The definitions become arithmetic.** With $H = 1$ the value of the policy “pull arm i ” (Defini-
40 tion ??) is the arm mean, $f_t^\pi = \mu_i$. If every arm is representable, the within-capacity optimum
41 (Definition ??) is $f_t^{K_L} = \mu^* = \frac{1}{2} + \epsilon$ and the structural gap vanishes, $\Delta_t^{\text{str}} = 0$. The learner
42 $\mathcal{L}$ selects the played arm a_t ; the examiner $\mathcal{E}$ (Definition ??) reads its two numbers off the same
43 counters,

$$V_t^{\text{hi}} = \max_i (\hat{\mu}_i + c_i), \quad V_t^{\text{lo}} = \hat{\mu}_{a_t} - c_{a_t},$$

44 an optimistic ceiling over *all* arms and a pessimistic floor on the *played* arm, with inductive bias $\mathcal{M}_E$
45 equal to the concentration rule that lets c_i shrink with data. The examiner regret (Definition ??) is
46 then the confidence width,

$$\rho_t = V_t^{\text{hi}} - V_t^{\text{lo}} = \max_i (\hat{\mu}_i + c_i) - (\hat{\mu}_{a_t} - c_{a_t}),$$

47 which uses only counts and empirical means, never μ or μ^* . These are not new objects: $\sum_t (f_t^{K_L} -$
48 $f_t^{\pi_t}) = \sum_t (\mu^* - \mu_{a_t})$ is the classical pseudo-regret and $\sum_t \rho_t = O(\sqrt{NT \log T})$ is the standard
49 UCB confidence-width sum (??).

50 **Where honesty lives.** Call the examiner *sound* if its floor is a true floor, $V_t^{\text{lo}} \leq f_t^{\pi_t}$, and its ceiling
51 a true ceiling, $V_t^{\text{hi}} \geq f_t^{K_L} - \delta_t$; by Hoeffding both hold with high probability, with slack δ_t the
52 failure margin that vanishes as counts grow. The mechanism is one line: an arm never pulled keeps
53 c_i at its maximum, so its optimistic index stays high and the ceiling V_t^{hi} cannot fall. The examiner
54 is structurally forbidden from reporting “no room to improve” about an arm it has not inspected; a
55 dishonest examiner is exactly one that discards this and assumes the unseen arm is ordinary.

56 **Three behaviours (Figure ??).** First, a self-report alone certifies nothing: a trivial examiner re-
57 porting $\rho_t \equiv 0$ pairs equally with a good (UCB) learner and a bad (fixed-arm) learner, identical
58 reports but opposite true regret (panel A). Second, when the world is coverable ($N = 5$), a sound
59 $\sum_t \rho_t$ stays above the true within-capacity regret throughout, a valid if loose certificate of what
60 remains removable (panel B). Third, when the world outruns the horizon ($N > T$) and the special
61 arm is placed out of reach, a sound examiner honestly keeps ρ_t large, signalling uncovered ground,
62 while a biased examiner that assumes a shared mean collapses ρ_t toward zero and reports a false
63 “done” even as true regret climbs at rate ϵ (panel C). This last case is the price the position pays in a
64 big world: an honest, vanishing self-report is not free.

65 **How to use it.** The bandit shows the three intended uses without any external optimum. As a *mon-*
66 *itor*, ρ_t is an online read of self-assessed room to improve; as a *comparison metric*, two algorithms
67 on one stream are ranked by which ρ_t shrinks faster toward its capacity-constrained optimum; and
68 as an *audit*, a low ρ_t is trustworthy exactly on the covered region, with the big-world case above
69 marking when no audit is possible and apparent convergence should instead trigger exploration or
70 external testing.

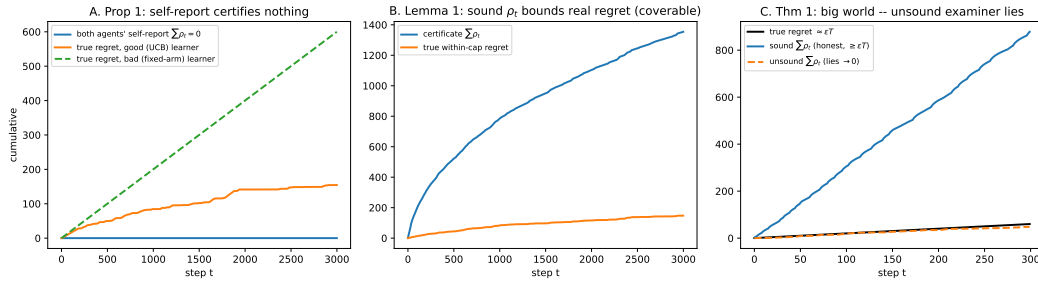


Figure 1: The examiner in a bandit. **(A)** Two agents with the identical self-report $\sum_t \rho_t = 0$ differ arbitrarily in true regret. **(B)** In a coverable world the sound certificate $\sum_t \rho_t$ upper-bounds the true within-capacity regret. **(C)** When the world outruns the horizon, a sound examiner keeps ρ_t large (honest), while a biased examiner drives ρ_t below the true regret (unearned confidence).